

Institutional Response to Public Testimony on Redistricting Communities of Interest

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Methods

Data Sources

The GeoNames database compiles place names alongside their corresponding point or polygon geometries at all administrative levels around the world (“GeoNames Gazetteer Extract Files” 2026). Pennsylvania GeoNames locations, alongside school district names from the **tigris** package in R (Walker 2025), serve as the building block unit of each commenter’s community of interest. Public testimony data was gathered from the Redistricting Data Hub, which has compiled comments from the 2020 redistricting cycle for 14 individual states (Redistricting Data Hub 2022). Tabula, a software capable of converting PDFs to csv files, was used to process comments about Pennsylvania state house boundaries and commenter communities of interest. Comments were combined with a specified community name each commenter was asked to provide.

Ellmer Pipeline

The **ellmer** package in R was used to interact with off-the-shelf LLM chat features and extract structured data from public testimony comments for Pennsylvania (Wickham et al. 2025). Barrie, Palmer, and Spirling (2025) recommend the use of open-weight Large Language Models (LLMs), which allow for more consistent replication of coding and classification tasks compared to the more fragile but advanced proprietary frontier LLMs. Therefore, we make use of an OpenRouter “GPT-4o” model combined with an **ellmer** schema to gather comment sentiment, comment clarity, and every specific location mentioned within a comment. Specific locations were defined by a schema of their own consisting of the location name, its administrative level, a binary indicator of whether the commenter mentions some or all of the location, additional descriptors or mentioned subareas, an estimate of the Pennsylvania counties corresponding to the location, and the community boundary group of the location based on the commenter’s specification.

Community boundary groups illustrated commenter distinctions between communities without making assumptions about the commenter’s location of residence. For instance, if a commenter stated that State College and Boalsburg, two adjacent municipalities in Central Pennsylvania, be kept together, but Bellefonte, the nearby county seat of Centre County, be placed in a separate congressional district, State College and Boalsburg would be given a group assignment of 1 (provided one of the locations was the first one mentioned in the comment), and Bellefonte would be given a group assignment of 2. Comment sentiment and clarity are both scaled to range from 0 to 1, with 0 representing negative sentiment and unclear comments while 1 represents positive sentiment and clear comments. The date of each comment was available from the original dataset, and comment character length was recorded as well.

GeoNames Matching

For each mentioned location in every comment, potential GeoNames location matches were filtered to the “P”, “L” and “A” feature class designations (see <http://www.geonames.org/export/codes.html>) as well as

the corresponding set of identified counties for that location. This set of potential matches was combined with all of the available Pennsylvania school district names as well as any alternate names for each GeoNames location and all GeoNames locations without a county level designation, meaning the location is larger than a single county. This combined list was referenced against the comment's specific location mentions gathered by **ellmer** using the **stringdist** package (Van Der Loo 2014). Matches with the lowest jaccard distances from **stringdist** were assumed to be the GeoNames equivalent.

References

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- Walker, Kyle. 2025. *Tigris: Load Census TIGER/Line Shapefiles*. <https://doi.org/10.32614/CRAN.package.tigris>.
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